FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad

Data Analysis and Visualization

Lab Manual 3

Lab Learning Outcomes:

- Build the same dataset as multiple chart types and judge which best fits a given analytical task.
- Recognize and reproduce common misleading techniques (e.g., truncated axes, dual axes) and explain the distortion they cause.
- Detect and fix a color-accessibility issue in a chart using a colorblindness simulator.
- Justify chart, color, and design choices in writing with specific technical reasoning.

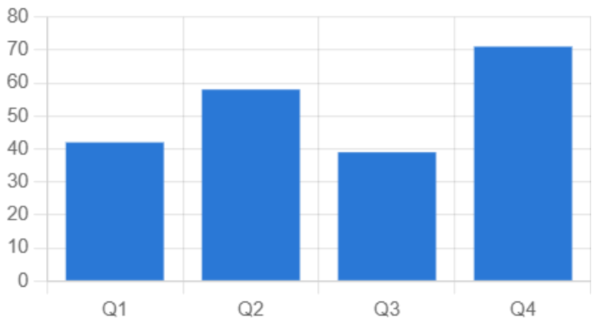
Submission Requirements:

- Submit notebook (.ipynb) + cleaned dataset (.csv) on both GCR and GitHub ,incomplete without both
- GitHub: push into your section repo, inside the week's folder, in your own subfolder named RollNumber_Section (e.g., 24F1234_5A)
- GCR: file naming convention RollNumber_Section_LabXX (e.g., 24F1234)
- Keep your GitHub subfolders updated weekly ,a complete, organized profile is checked at the Capstone stage (Week 15–16)

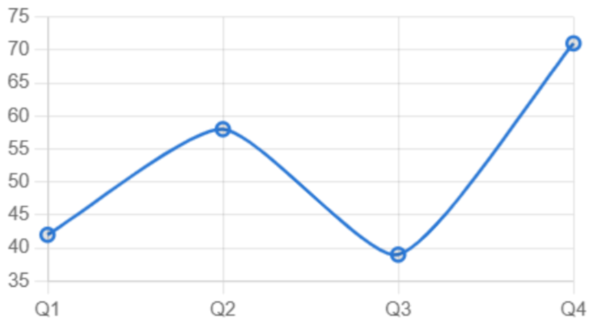
1. Same data, multiple chart types
Take one dataset (e.g. monthly sales) and render it as a bar chart, line chart, and pie chart. This exposes what each form is actually good at.

Example:

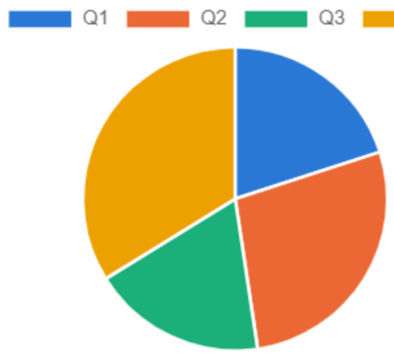
Bar — good for comparing discrete categories



Line — good for trend over time



Pie — weak here: only 4 slices, differences hard to judge, no time signal



2. Effectiveness & appropriateness

Task	Best form	Weak choice
Compare magnitudes across categories	Bar/column	Pie, radar
Show trend over time	Line, area	Bar (for many points)
Part-to-whole, few categories	Stacked bar, pie (≤4 slices)	3D pie
Correlation between two variables	Scatter	Line
Distribution/spread	Histogram, box plot	Bar of averages
Precise value comparison	Bar, table	Pie, bubble

3. Accessibility

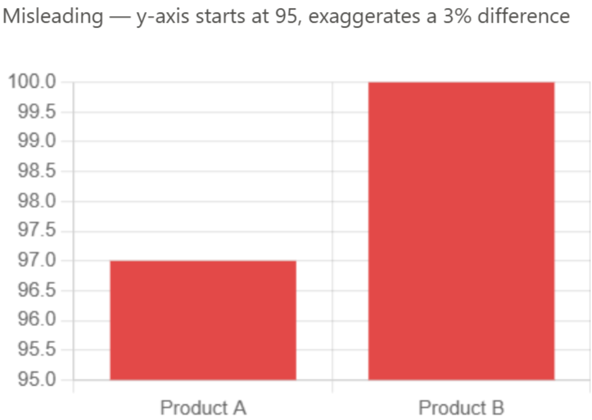
- **Color blindness**: ~8% of men have red-green deficiency. Never encode meaning with red vs. green alone. Pair color with shape, pattern, or direct labels.
- **Contrast**: text and gridlines need sufficient contrast against background (WCAG AA: 4.5:1 for text).
- **Alt text / screen readers**: every chart needs a text summary of the data, not just a visual ,role="img" + aria-label, or a data table alternative.
- **Don't rely on color alone**: dash patterns for lines, textures/hatching for bars, distinct icons or markers.

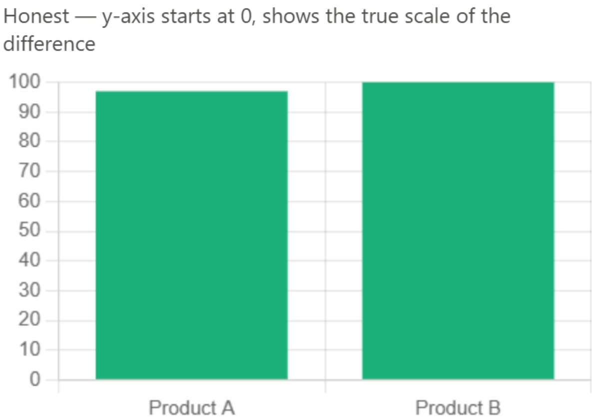
4. Color selection

- **Categorical** (distinct groups, no order): use a fixed set of distinguishable hues, cap at ~8, never a rainbow gradient for unordered categories.
- **Sequential** (low→high magnitude): one hue, light to dark.

- **Diverging** (above/below a midpoint): two hues meeting at a neutral gray, not another hue.
- Avoid using color to imply an order that doesn't exist, and avoid saturated red/green pairs for the reason above.

5. Misleading visualizations





Other classic misleading techniques:

- **Dual y-axes** ,two unrelated scales overlaid to imply a correlation that isn't there.
- **3D effects** ,distort area/volume perception, especially in pie and bar charts.
- **Cherry-picked time windows** ,showing only the range that supports the narrative.
- **Inverted or non-linear axes** ,reversing an axis to make a decline look like growth.
- **Inconsistent bin sizes** in histograms, or cumulative charts that always look like growth.
- **Area/bubble size scaled by radius instead of area** ,visually overstates differences (a value 2x as large gets 4x the visual area).

Lab Tasks:

1. You will be given the total monthly sales already summarized by month. Plot this as a bar chart, a line chart, and a pie chart: *"Which chart shows the trend best, and which one hides it? Why?"*
2. For each statement below, pick the chart type, build it, and justify in 1-2 sentences why it fits and why the "weak choice" would mislead or underperform:

   "Compare total sales across the top 5 sub-categories." (You will be given the top 5 sub-categories already totaled.)

   "Show the revenue trend over time." (Reuse the monthly sales data from Task 1.)

   "Show the distribution of order-line sales values." (Use the raw sales column directly — no summarizing needed.)

   "Show the relationship between discount level and profit." (Use the raw discount and profit columns directly — no summarizing needed.)

3. You will be given: total sales by region, sales by region-and-category, and total profit by month.
   (a) Recolor the region sales chart with a proper categorical palette.
   (b) Recolor the region-by-category data as a heatmap using a sequential palette.
   (c) Build a diverging-color chart of monthly profit, since it crosses zero.
4. Write a function that computes the WCAG contrast ratio between two RGB colors, and test it against the colors used in Task 3

   (a) flag any that fail the 4.5:1 AA threshold.

   (b) You will be given profit by sub-category. Build a chart that encodes loss vs. gain using red and green only. Run it through a provided colorblind-simulation snippet and screenshot before/after. Then fix the chart using hatching or marker shape so the distinction survives the simulation, and write one line of alt text describing the chart.

5. Using the region-sales and monthly-sales/monthly-profit data from earlier tasks, produce at least 3 of the following, each with a one-line caption naming the exact distortion mechanism:

   - A truncated y-axis on the region sales chart
   - A dual-axis chart overlaying two unrelated measures to imply a fake correlation
   - A cherry-picked date window that hides the surrounding trend
   - An inverted y-axis on the monthly profit or sales line
   - A bubble chart where circle radius (not area) scales with value

6. Pick one real analytical question about the dataset (e.g., *"Is discounting hurting profit?"* using the raw discount/profit columns — no summarizing required). Submit:

- The honest, appropriately-chosen chart answering it.
- A deliberately misleading version of the same question.
- A short written justification covering chart type choice, color choice, and an accessibility check.